BRI Resilience Function Specification

Purpose

This specification defines a practical "BRI resilience function" for flood-related physical mitigation that can be implemented alongside an existing hazard damage function which currently ignores mitigation measures. The function does **not** replace the existing damage model. Instead, it generates a resilience score and a damage-scaling modifier that converts baseline hazard damage into a mitigation-adjusted loss-given-hazard estimate.[cite:81] [cite:61] [cite:118]

The design intentionally borrows the logic of structured resilience checklists and prioritised measures, but it does **not** use the official Building Resilience Index weakest-link rollup. BRI itself is measure-based, relative, and not a direct catastrophe simulation model, which makes it a suitable conceptual template for a practical implementation layer over an existing damage curve.[cite:81]

Design principles

The implementation should follow five principles:

- Use a continuous 1-100 score rather than a letter grade so the result can be consumed directly by pricing, underwriting, or portfolio models.[cite:61] [cite:116]
- Weight measures according to their expected contribution to reducing conditional flood damage, not merely their presence in a compliance checklist.[cite:110] [cite:113] [cite:118]
- Preserve non-binary measure quality wherever practical because several relevant resilience attributes are naturally continuous or ordinal rather than yes/no.[cite:81] [cite:113]
- Separate physical mitigation from data-quality or verification confidence; confidence can widen uncertainty bands but should not directly alter physical damage if the measure is actually present.[cite:81]
- Apply the resilience output as a multiplier to an existing hazard damage function, rather than replacing hazard physics or baseline vulnerability assumptions.[cite:118]

Functional architecture

The implementation should produce four outputs for each asset-hazard evaluation:

1. `flood_resilience_score_raw`: a weighted score from 0 to 100 before any caps or hazard-specific adjustment.
2. `flood_resilience_score_final`: a weighted score from 1 to 100 after any soft caps or hazard-specific logic.
3. `flood_damage_modifier`: a multiplier applied to the existing baseline damage function.
4. `flood_resilience_confidence`: a separate confidence indicator based on evidence quality and verification completeness.[cite:81] [cite:118]

The adjusted loss-given-hazard calculation is:

$$\mathrm{LGH}_{adj} = \mathrm{BaselineDamage}(h, x) \times M(S, r)$$

where:

- h is hazard intensity,
- x is the existing vulnerability input set,
- S is the final resilience score,
- r is flood regime or subtype, such as pluvial, fluvial, or coastal.[cite:61] [cite:118]

Measure model

Response types

Measures should be stored in one of three forms depending on what is actually available from assessors or source systems.[cite:81]

Response type	Description	Examples
Continuous	Numeric physical quantity mapped into a 0-1 compliance factor	floor elevation, critical systems elevation, percent permeable area
Ordinal	Discrete quality band mapped into a 0-1 compliance factor	retainage capacity, wet-floodproofing quality, materials resilience
Binary	True/false presence flag mapped into 0 or 1	independent backflow valve present, known historic watercourse encroachment absent

Compliance factors

Each measure is converted into a compliance factor c_i between 0 and 1. Recommended generic ordinal mapping is:[cite:116]

Assessor response	Compliance factor
None / adverse	0.00
Minimal	0.25
Partial	0.50
Strong	0.75
Best available / fully aligned	1.00

Core flood resilience checklist

The checklist below is designed specifically for use with a baseline damage function. It emphasizes measures most likely to affect flood depth at the structure, exposure of vulnerable components, and severity of post-entry damage.[cite:61] [cite:110] [cite:113] [cite:118]

Measure code	Measure name	Response structure	Weight	Notes
`lowest_occupied_floor_elevation`	Lowest occupied floor elevation above local water reference	Continuous/banded	20	Highest-impact control on water reaching occupied or vulnerable space.[cite:81][cite:61]
`critical_systems_elevation`	Elevation of electrical and critical M&E systems above local water reference	Continuous/banded	15	Strong effect on severe loss and downtime.[cite:81][cite:113]
`site_drainage_topography`	Site grading and drainage away from building	Ordinal or percentage-derived	10	Stronger for pluvial/local flood contexts.[cite:81]
`onsite_retainage_capacity`	Onsite stormwater retainage / attenuation standard	Ordinal	10	BRI distinguishes 100-year and 200-year capacities.[cite:81]
`permeable_surface_share`	Share of external ground area that is permeable	Continuous percent	6	Reduces runoff generation around asset.[cite:81]
`backflow_protection`	Coverage of backflow prevention on vulnerable lines	Ordinal	6	Important in urban/pluvial contexts.[cite:81]
`lower_level_flood_compatible_design`	Flood-compatible or wet-floodproofed lower exposed level	Ordinal	8	Limits damage when lower level is expected to flood.[cite:81][cite:110]
`water_resistant_materials`	Use of water-resistant finishes/materials in exposed lower levels	Ordinal	8	Reduces repair severity after inundation.[cite:81][cite:110]
`debris_impact_resistance`	Resistance of exposed envelope/openings to debris impact	Ordinal	5	More material in fast-flow/debris environments.[cite:81]

Measure code	Measure name	Response structure	Weight	Notes
`roof_drainage_standard`	Roof drainage design and maintenance standard	Ordinal	4	Helps during intense rainfall and roof water concentration. [cite:81]
`roof_membrane_openings_seal`	Quality of roof membrane and opening seals	Ordinal	3	Secondary for true flooding but relevant to water ingress. [cite:81]
`sump_pumps_backup_power`	Sump pump and resilient backup power availability	Ordinal	3	Especially relevant for basements and urban flooding. [cite:81]
`historic_water_area_flag`	Site history of natural water area / adverse water legacy	Binary or ternary adverse flag	2	Treated as a vulnerability screen, not a core mitigation. [cite:81]

Total base weight = 100.

Response mappings

1. Lowest occupied floor elevation

This field should store the measured elevation margin in meters above the relevant local water reference. The local water reference should be defined consistently by hazard regime: local grade/drainage low point for pluvial contexts, design flood level for fluvial contexts, and relevant surge/tidal flood level for coastal contexts.

Suggested compliance mapping :[cite:81]

Elevation margin	Compliance factor
Less than 0 m	0.00
0 to less than 1 m	0.20
1 to less than 3 m	0.50
3 to less than 5 m	0.75
5 to less than 6 m	0.90
6 m or more	1.00

2. Critical systems elevation

This field should represent the elevation margin of the most vulnerable critical equipment set, including electrical panels, switchgear, telecom hubs, pumps, controls, and key M&E equipment. [cite:81] [cite:113]

Elevation margin	Compliance factor
Less than 0 m	0.00
0 to less than 1 m	0.25
1 to less than 3 m	0.60
3 to less than 5 m	0.85
5 m or more	1.00

3. Site drainage topography

This should not be binary. It can be derived from percent of perimeter draining away from the building or entered as an assessed ordinal quality.

Condition	Compliance factor
Drainage concentrates toward building / poor grading	0.00
Mixed grading, limited drainage benefit	0.25
Moderate grading away, some pinch points	0.50
Good grading away from most exposed perimeter	0.75
Strong site drainage away with no obvious ponding traps	1.00

4. Onsite retainage capacity

Suggested mapping aligned to BRI-style thresholds:[cite:81]

Condition	Compliance factor
None / clearly undersized	0.00
Limited attenuation below 100-year equivalent	0.25
Approximately 100-year design standard	0.60
Better than 100-year but below 200-year	0.80
200-year standard or engineered equivalent	1.00

5. Permeable surface share

This field should capture the share of relevant external area that is permeable.

Permeable share	Compliance factor
Less than 10%	0.00

Permeable share	Compliance factor
10% to less than 30%	0.25
30% to less than 50%	0.50
50% to less than 80%	0.75
80% or more	1.00

6. Backflow protection

Condition	Compliance factor
None	0.00
Limited branch protection only	0.25
Main vulnerable lines protected	0.50
Broad protection with some redundancy	0.75
All vulnerable lines protected and maintained	1.00

7. Lower level flood-compatible design

Condition	Compliance factor
Vulnerable occupied lower level, no flood-compatible design	0.00
Some flood-compatible detailing only	0.25
Partially flood-compatible lower level	0.50
Predominantly flood-compatible lower level with limited vulnerable uses	0.75
Robust wet-floodproofed or sacrificial lower level, non-critical use	1.00

8. Water-resistant materials

Condition	Compliance factor
Standard vulnerable materials	0.00
Limited resilient materials in small areas	0.25
Partial resilient fit-out in exposed areas	0.50
Extensive resilient materials in exposed lower levels	0.75
Robust water-resistant material strategy throughout exposed levels	1.00

9. Debris impact resistance

Condition	Compliance factor
None	0.00
Minimal local strengthening	0.25

Condition	Compliance factor
Partial design for exposed openings/walls	0.50
Strong design response for key exposures	0.75
Designed and verified for expected debris conditions	1.00

10. Roof drainage standard

Condition	Compliance factor
Below standard / poor maintenance	0.00
Meets basic minimum only	0.25
Moderate enhanced design or maintenance	0.50
Strong above-minimum roof drainage provision	0.75
High-capacity and well-maintained system	1.00

11. Roof membrane and opening seals

Condition	Compliance factor
Poor / known leakage vulnerabilities	0.00
Limited upgrade to membrane or openings only	0.25
Moderate water-ingress resistance	0.50
Strong membrane and opening-seal strategy	0.75
Robust, maintained, and verified envelope sealing	1.00

12. Sump pumps and backup power

Condition	Compliance factor
No sump and no resilient power	0.00
Sump only or unreliable backup	0.25
Sump with limited backup/runtime	0.50
Good pumping setup with resilient backup	0.75
Robust pumping redundancy and reliable backup power	1.00

13. Historic water area flag

This measure is best treated as a site-vulnerability flag rather than a mitigation measure.

Condition	Compliance factor
Known adverse history / encroachment on natural water area	0.00
Uncertain / incomplete evidence	0.50

Condition	Compliance factor
Clear absence of adverse condition	1.00

Raw score calculation

The raw resilience score is:

$$S_{raw} = 100 \times \frac{\sum_i w_i c_i}{100}$$

Since the weights sum to 100, the implementation is numerically equivalent to:

$$S_{raw} = \sum_i w_i c_i$$

The result should be clipped to the interval [0, 100].

Hazard-regime weighting adjustment

A single universal flood score is useful, but the score should optionally adapt to flood subtype because some measures have different relevance under pluvial, fluvial, and coastal conditions. [cite:61] [cite:81] [cite:118]

The recommended implementation is to maintain the same checklist but apply a regime-specific weight multiplier before normalization.

Suggested regime multipliers

Measure code	Pluvial	Fluvial	Coastal
`lowest_occupied_floor_elevation`	1.0	1.2	1.2
`critical_systems_elevation`	1.0	1.1	1.1
`site_drainage_topography`	1.3	0.9	0.8
`onsite_retainage_capacity`	1.3	0.9	0.8
`permeable_surface_share`	1.3	0.8	0.7
`backflow_protection`	1.2	1.0	0.9
`lower_level_flood_compatible_design`	0.9	1.2	1.2
`water_resistant_materials`	0.9	1.1	1.1
`debris_impact_resistance`	0.8	1.0	1.2
`roof_drainage_standard`	1.1	0.9	0.9
`roof_membrane_openings_seal`	1.0	1.0	1.0
`sump_pumps_backup_power`	1.2	1.0	0.9
`historic_water_area_flag`	1.0	1.0	1.0

For a selected regime r, use adjusted weights:

$$\tilde{w}_{i,r} = w_i \times m_{i,r}$$

and compute:

$$S_{raw,r} = 100 \times \frac{\sum_i \tilde{w}_{i,r} c_i}{\sum_i \tilde{w}_{i,r}}$$

This preserves a 0-100 scale while making the score more physically relevant to the flood type. [cite:61] [cite:118]

Soft caps

Soft caps are recommended to prevent unrealistic scores when the most damage-relevant controls are poor but many secondary measures are present.

Suggested caps:

Condition	Cap on final score
Lowest occupied floor elevation compliance = 0.00	40
Critical systems elevation compliance = 0.00	60
Both retainage compliance and site drainage compliance are 0.00 in pluvial regime	55
Lower level flood-compatible design = 0.00 and water-resistant materials = 0.00 in fluvial/coastal regime	65

Final score is:

$$S_{final} = \min(S_{raw,r}, \text{all applicable caps})$$

If no caps apply, then $S_{final} = S_{raw,r}$.

To avoid a zero score in downstream systems that expect a positive value, implementers may clip the final score to the interval [1, 100].

Damage modifier

The resilience function should scale the existing damage function rather than replace it.[cite:118]

Minimum recommended form

A bounded linear modifier is the simplest operational starting point:

$$M(S) = 1 - \alpha \times \frac{S - 1}{99}$$

where α is a calibration parameter controlling maximum mitigation effect. Recommended initial value: $\alpha = 0.60$.[cite:118]

This gives approximate behavior:

Final resilience score	Damage modifier at $\alpha = 0.60$
1	1.00
25	0.85
50	0.70
75	0.55
100	0.40

Regime-sensitive modifier

If the tech team prefers more realism, the same score can map to a regime-specific multiplier:

$$M(S, r) = 1 - \alpha_r \times \frac{S - 1}{99}$$

with example starting parameters:

Regime	Suggested α_r
Pluvial	0.65
Fluvial	0.60
Coastal	0.50

This reflects that local drainage and local-property measures often have stronger leverage in pluvial flooding than in deep coastal inundation, where residual losses remain high even for well-mitigated properties.[cite:61] [cite:118]

Confidence score

Confidence should be calculated separately from physical mitigation. This can be used to widen model uncertainty, trigger manual review, or downgrade trust in automated decisions.[cite:81]

Suggested confidence inputs:

Confidence factor	Suggested scoring
Independent verification present	0 or 1
Evidence quality	0 to 1
Date of last inspection	0 to 1
Completeness of measure fields	0 to 1

Example confidence score:

$$C = 100 \times \frac{0.35v + 0.25e + 0.20d + 0.20k}{1.00}$$

where:

- v = verification completeness,

- e = evidence quality,
- d = recency factor,
- k = field completeness.[cite:81]

This confidence score should not directly alter the mean damage modifier unless the business explicitly wants a conservative penalty framework. A better default is to use confidence to widen or narrow uncertainty around modeled loss.[cite:81] [cite:118]

Pseudocode

```
def compute_compliance(measure_code, raw_value):
    # implement table-driven mappings for each measure
    return c_i  # value in [0, 1]


def compute_regime_adjusted_score(measures, regime):
    base_weights = {
        "lowest_occupied_floor_elevation": 20,
        "critical_systems_elevation": 15,
        "site_drainage_topography": 10,
        "onsite_retainage_capacity": 10,
        "permeable_surface_share": 6,
        "backflow_protection": 6,
        "lower_level_flood_compatible_design": 8,
        "water_resistant_materials": 8,
        "debris_impact_resistance": 5,
        "roof_drainage_standard": 4,
        "roof_membrane_openings_seal": 3,
        "sump_pumps_backup_power": 3,
        "historic_water_area_flag": 2,
    }

    regime_multipliers = {...}  # as specified above

    weighted_sum = 0.0
    weight_total = 0.0
    compliance = {}

    for code, base_w in base_weights.items():
        c = compute_compliance(code, measures.get(code))
        compliance[code] = c
        w = base_w * regime_multipliers[regime][code]
        weighted_sum += w * c
        weight_total += w

    s_raw = 100.0 * weighted_sum / weight_total if weight_total else 0.0

    caps = [100.0]
    if compliance["lowest_occupied_floor_elevation"] == 0.0:
        caps.append(40.0)
    if compliance["critical_systems_elevation"] == 0.0:
        caps.append(60.0)
    if regime == "pluvial" and compliance["onsite_retainage_capacity"] == 0.0 and compli
        caps.append(55.0)
    if regime in {"fluvial", "coastal"} and compliance["lower_level_flood_compatible_des
        caps.append(65.0)
```

```
    s_final = max(1.0, min(s_raw, min(caps)))
    return s_raw, s_final, compliance


def damage_modifier(score_final, regime):
    alpha = {"pluvial": 0.65, "fluvial": 0.60, "coastal": 0.50}[regime]
    return 1.0 - alpha * ((score_final - 1.0) / 99.0)


def adjusted_loss_given_hazard(baseline_damage, measures, regime):
    s_raw, s_final, compliance = compute_regime_adjusted_score(measures, regime)
    modifier = damage_modifier(s_final, regime)
    return {
        "flood_resilience_score_raw": s_raw,
        "flood_resilience_score_final": s_final,
        "flood_damage_modifier": modifier,
        "loss_given_hazard_adjusted": baseline_damage * modifier,
        "compliance_factors": compliance,
    }
```

Implementation guidance

- Store raw physical inputs where possible rather than only storing pre-banded values; this allows later recalibration without changing the source schema.[cite:113] [cite:118]
- Keep the mapping tables externalized in configuration rather than hard-coded so model governance and recalibration are easier.[cite:118]
- Log both the raw and final score, plus each component compliance factor, so model outputs are explainable to underwriting, audit, and engineering stakeholders.[cite:81] [cite:118]
- Treat missing values explicitly. Missing should not silently behave as good compliance. The recommended default is either `None -> unknown` with imputation rules, or conservative mapping to 0.25 plus a confidence penalty, depending on business preference.[cite:81]
- Calibrate the α parameters and any score caps empirically when claims or engineering loss data become available. The proposed values are suitable for first implementation, not final scientific calibration.[cite:118]

Minimum data contract

The implementation should support the following minimum measure payload for each property-flood evaluation:

Field	Type
`lowest_occupied_floor_elevation_m`	float
`critical_systems_elevation_m`	float
`site_drainage_topography_level`	enum or numeric
`onsite_retainage_capacity_level`	enum
`permeable_surface_share_pct`	float
`backflow_protection_level`	enum

Field	Type
`lower_level_flood_compatible_design_level`	enum
`water_resistant_materials_level`	enum
`debris_impact_resistance_level`	enum
`roof_drainage_standard_level`	enum
`roof_membrane_openings_seal_level`	enum
`sump_pumps_backup_power_level`	enum
`historic_water_area_flag`	enum or boolean
`flood_regime`	enum: pluvial, fluvial, coastal
`baseline_damage`	float

Deliverable summary

The recommended implementation is a table-driven resilience layer that computes a flood resilience score from weighted non-binary measures, applies hazard-regime logic, enforces soft caps for the most damage-critical weaknesses, and converts the result into a multiplicative modifier on the existing baseline damage function.[cite:61] [cite:81] [cite:118]

This approach is simple enough for immediate implementation, retains alignment with BRI-style measure logic, and is materially more suitable for a loss-given-hazard framework than a weakest-link letter rating.[cite:81] [cite:118]